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This week, we toss out the idea that digitization and digital creations are simply archived copies of the documents they seek to preserve/present.
Rather, choices are made at each step of the process of digitization/digital creation that shape how the final product exists and the impact it has; the digital methods and platforms used impact how something can exist and work upon the world.

Digital works are particular versions of the original text with particular differences, not a replacement.

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Notes on digitization from class:

We reject:

- idea that digitization is preservation, and we can just throw everything else out
    - if a scan is the only copy, not copyrighted, transformative
    - why ai orgs buying all books and destroying them after a scan
- idea that scan in super high resolution is an archival copy
    - many do capture almost all of the visual data on the document... but still there exist choices that differentiate the digital from the original
    - and what of the material data that cannot be digitized?
    - even the first decision to scan something depends on the physicality of the object and and on the technology available (the digital object)
    - non-choices like the amount of light that can be used for a given tool also shape the final digital work
      if we made the digital physical again... is it the same as it was? Do the choices we make when we digitize something not affect what is possible for the next digitization

- metadata can be explained in a number of way - a set of terms in metadata that have been hammered out through conferences... the terms of your metadata decides how your object can speak and interact (be found etc)
    - what about being remixable though?

Notes on readings:

"Q i-jtb the Raven" Taking Dirty OCR Seriously Notes

- text mining relies not on texts themselves, but on texts as remediated by mass digitization (labour intensive, tech intensive, economically determined process)
- OCR-derived text underlies our searches
- Kirschenbaum and Werner call for the "perspectives of book historians … on the large-scale digitization efforts underway at such places such as Google Books, the Internet Archive, HathiTrust, Gallica, and other institutions that are actively aiming to make print resources available as digital objects."
    - uneven, unclear metadata quality
    - 2D representation, flat
    - single-copy digitization (I wonder if this sort of canonizes one particular version, without necessarily a tremendous amount of thought as to why)
      -> How can we digitize bibliographically to account for "sources, technologies, and social realities of their creation" so people can discuss and critique?
    - Digitization is part of that bibliography, rather than separate
- the e-book, Galey shows, is "a complex transaction of electronic signals, material inscriptions, and cultural codes" which we interpret through "layers and layers of digital tools and interfaces.
- often do not *see* the digital object, but surrogate of the print object from which it is derived. Why are we citing and discussing digital objects as if discovered through analog means? removed role of the web interfaces, human labor, and algorithmic search technologies that shape what can(t) find
    - ""We still tend to think of computers as physical machines rather than the software that makes them go … we think of the iPad, but not the Safari browser that comes on every iPad"
- Errorful OCR influences influences our research, inhibiting, for instance, comprehensive search.
    - correcting costs time and money
- copes letters and words from form rather than by sense
- while human compositors and correctors typically did recognize the words they composed, their processes could be more automatic than we might countenance

- Mak describes the layers of a digitization's image, text data, metadata, and interface, "as palimpsests" altering each other, "a particular synthesis of traditional and emergent technologies" that are "challenging to locate for scholarly analysis."
- "OCR-derived text so rarely *can* stand in for the texts from which it derives" surrogacy is myth
- such models also flatten our understanding of digitizations, encouraging misrecognitions that stymie both computational and bibliographic thinking. When the "It" of an interface refers only to a print edition, it reinforces codexical thinking in the midst of a digital edition rich with machine reading possibilities
- If the digitized version of a text is a new, clearly-distinct edition, then it cannot stand as "one representative copy" for all the physical copies of its print original.52 A "'surrogate first' policy that restricts access to print copy" makes no sense when the digital edition cannot be considered a surrogate... but argues surrogacy not a good way of understanding anywys. This a new edition impacted by its method of digitization, sources, the choices of digitizers, and context of production
- funding and time limitations lead to digitization errors... shaped by funding and thereby by the economic and politcal conditions in which produced. Some places' project sunderfunded and others not at all
- '" A historical object in a large scale digital archive evidences not only the time period of its print original and its online present. It witnesses also to cultural and scholarly modes of transmission that bridge the period of the historical object's creation and our experience of its digitized editions."
- bringing out the bibliographies of sources *forefronts their createdness*
- not a technical challenge, but a challenge of imagination.

‘Q i-jtb the Raven’: Taking Dirty OCR Seriously Blog Post

- provides a bibliography of own text, showing its evolution from a blog post, and what it does not capture due to time restrictions

Sparrow, T., Bain, K., Kimber, M. and Wilson, A.S. 2024 Visualising Heritage

- completed for Grand Junction Railway (GJR), Stephenson's Roundhouse (the world's first Railway Roundhouse), and Curzon (Principal) Building
  - used scalar approaches to terrestrial and airborne capture, from the object scale to landscapes, including complex cityscapes
  - long-range terrestrial laser scans obtained using a FARO X330 laser scanner
  - mobile mapping using a GeoSLAM Zeb Horizon
  - a total of 667 images in RAW format were captured for the Grand Junction Railway Booking office, with a further 7,565 images captured for the Roundhouse

    - Terrestrial scan data and structure-from-motion photogrammetry were also used during the excavation of the former Park Street Cemetery to document brick-lined vaults and exemplars of earthen graves
- allows for an analysis of changes over time as compare original plans to scans of buildings
- methods of creating engaging models and visuals to inform and influence wider public and city/national projects (HS2 Phase One professionals here). richly interactive
    - Stephenson's roundhouse reburied to protect archeological remains
- scope restricted by funding and time
- data deposited with the Archaeology Data Service in York

Handwriting Recognition Roundup + Reflections

- has gemini 3.0 solved handwriting recognition?
- "doing this better than the best human-trained models and even better than actual humans trained in paleography at a graduate level."
  -> I struggle with this article because I worry that it describes how Gemini is interacting with the text like how a human would, but AI doesnt *really* think
  ex. Humphries "describes testing Gemini's ability to read and reason through complex early modern ledger books" ... but is AI really "reasoning" when it follows a pattern?
  ex. "to argue mathematically and ethically that my prompt was factually incorrect."... is it really making an ethical argument? What do we mean by that?
- "It allows human beings to focus their time on the important, profound work of understanding another human being, rather than staring at a curlicue to grasp if it's an L or an I." This I can understand
- "But now that AI can read handwriting—presumably at scale—perhaps we should revive the dream of putting *everything* online. Not just the already-famous collections, not just materials in major languages, not just the stuff that's easy to process. Everything."
  WARNING WARNING BEEP BEEP BEEP! Maybe not everything should be transcribed and put online!
- "we still dont fully understand what it means to have AI as an intermediary between researchers and primary sources." -> would need to include this in bibliography of a project...
- but can we really disregard the ethical cost? Is this the position we should take?

Pre-tutorial class notes

- OCR is not something you do and it's done. Need to inject semantic info into it to be useful
    - comes out of mechanics aimed towards supported partially sited people
    - devices for scanning light, which can be used to scan for numbers -> how we got automatic cheque readers in 1950s

        - your computer learning the pattern of pixels for a 3 and for an E and lights up in their patterns
        - now can feed images... but images need to be cleaned up a lot to be read by the operating system
    - what we have now is a change not in what it's doing, but how much data it's familiar with and what it can do with it
      - Computer vision contest: a scholar finds that if she feeds it enough networks data, don't need to tell it in so much detail how to do to make something... it begins to know what to do
- network, input... can measure the distance between what system spits out as a cat and an actual image of cat.
    - measure the error and readjust things to get closer to reduce that error. Cycled over and over reducing the error, end up with a guess that is actually accepted by us. This is machine learning
    - can be used to image categorization and classification ... show it your target and it drops things through until it finds a close match
- for OCR, no longer a program with a find tuned model for everything, just need to feed it the info

Trying Out Generating an Ordered Data Set from an OCR Text File

Reading from PH:

- text representation of a data structure that lives in computer memory = dictionary or hash or associative array
    - easier to do programmatic analysis or manipulation with as already organized for data analysis
- want to use a string similarity algorithm to test probability that a string is a header, as regular expressions too unreliable with this messily composed strings
- "`lev()` function takes two strings as parameters and returns a number that indicates the 'string distance' between them, or, how many changes had to be made to turn the first string into the second."
    - like leveishene difference between log and bog is 1

Breaking it down for myself... I really need to know a bit more about what each function means to remotely understand

FIRST FUNCTION

def lev(seq1, seq2):

- `def` = creating a function.

- `lev` = the name we give function
- `seq1` and `seq2` = two pieces of text we want to compare.

rows = len(seq1)+1

- `len()` = count how many characters are in something.
-
    - 1 really just adds one
- rows is the name we give the number we get

**cols = len(seq2)+1

- here cols is the name we give the function, and does same thing for seq 2

distance = [[0]*cols for x in range(rows)]

= first function described in the article is bringing up the info and turning the info into columns. Has nested loops to calculate Levenshtein distance over and over

SECOND FUNCTION

returns arabic equivalent of roman numeral

REGULAR EXPRESSIONS

`re.compile()`

`.match()`, and `.search()`, `.findall()` and `.finditer()`

- `.match()` will only match at the **beginning** of a line, whereas `.search()` will match anywhere in the line **but then it stops**, it'll **only** return the first match it finds.
- `.findall()` and `.finditer()` will return **all** occurrences of the matched string;

Deploying it

- use iteratively, making new corrected versions of file each time
    - Every time you produce a text file, you should version it and duplicate it so that you can always return to it.
- One way to use the code snippets in section 2 might be to have all of them in a single file and comment out the bits that you don't want to run.
- maintain a single python module in which you define the dictionary at the top, along with your `import` statements and the assignment of global variables, followed by each of the four loops that will populate and then modify that dictionary

- we're going to be running it, opening the out1.txt, manually correcting errors, and then running it again. sometimes the script won't be able to find them because of other info on line
- chunk it up!
    - by page, by section... whatever makes sense
        - here a regex is used to find roman numerals that break up the sections
        - when the text fails
        1. hard to know where the page breaks even are in OCR, get thrown off.
        2. but can use headings that seperate each page, using regular expressions
    - using calculations of leviashian distance, we find headings with which to split up the text
        - "Once we've found, and fixed, all the roman numeral charter headings, then we can write out a new file with an easy-to-find-by-regex string, a 'slug,' for each charter in place of the bare roman numeral."
    - normalize markers by giving them their own lines - adding line breaks
- parse info and iteratively separate based on markers of organization you've chose, manually correcting errors and looping the functions to clean things up

Reflections

I approached this process attempting to understand what each line of code was doing, and realized after some hair-pulling that this detailed analysis wasn't the point of these tutorials. After muddling through the colab notebook Dr Graham sent us, I found that taking a step back to examine the big picture was the most useful part of the process for me.

This work is a continuously iterative process. As the OCR is often so disorganized and difficult for the system to read (and for regular expressions to make simple use out of), we require a continuous, looping, and often user-engaged process of organization. We make decisions about how the document should be chunked-up, about the most important information to preserve and what should be eliminated to improve clarity. The scripts we're using here are a tool for enacting our decisions, and for making text more searchable and usable for digital processes than it was before. Some of the work of correcting organizational issues and breaking things up falls to us, but the functions we're using allow us to identify where those errors are. Through our enactment of these processes, we are creating something new in the form of the python dictionary, and are making information/data more accessible for use generally.

Extra Notes

PPRINT MEANS PRETTY PRINT
PPRINT(CHARTERS)

means print the charters so easier to read

- then can use the html to print is as readable text. gives us a chunk of code for this and we can drop it into the browser to give us something readablE

DESCRIBING THE PATTERN OF THE WORK YOU WANT TO DO IS MORE IMPORTANT THAN WORKING OUT THE CODE

- a more important skill than knowimg the syntax precisely
- need to be able to read through code to understand flow and steps in process.. but if can describe pattern and what you are after in a way that makes sense, you can find success

DO NOT DO NOT DO NOT DO NOT USE THIS METHOD ON SENSITIVE INFORMATION WE ARE USING A SERVICE FROM A COMPANY CALLED GROQ (NOT ELON'S) TO ACCESS OCR

- THE GROQ CLOUD
  - A HARDWARE ARRANGEMENT THAT GIVES RESULTS FAST
- BUT YOU STILL GIVING INFO TO THIS ORG, SHOULD USE SOMETHING THAT JUST LIVES SOLELY ON YOUR COMPUTER